

Auteur: Tyas Hermans
 Studierichting: Informatica
 Oefeningenreeks 1D nr. 21.14

$$f(z) = z^4 - 5z^3 + 13z^2 - 19z + 10 = 0$$

$$f(1) = 1 - 5 + 13 - 19 + 10 = 0$$

$$f(z) = (z^3 - 4z^2 + 9z - 10)(z - 1)$$

$$f(2) = 8 - 16 + 18 - 10 = 0$$

1	1	-5	13	-19	10
		1	-4	9	-10
2	1	-4	9	-10	0
		2	-4	10	
	1	-2	5	0	

$$f(z) = (z^2 - 2z + 5)(z - 1)(z - 2)$$

$$D = 4 - 20 = -16$$

$$z_1 = \frac{2 + 4i}{2} = 1 + 2i$$

$$z_2 = \frac{2 - 4i}{2} = 1 - 2i$$

nulpunten: 1, 2, 1 + 2i, 1 - 2i

References